

Weather Station

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Embedded Systems Architectures



Introduction

What is this project?

Real-time data collection for temperature, humidity, pressure, wind speed & direction with autonomous operation with local storage and cloud connectivity.

Real-life Applications

Climate Awareness: Growing need for localized weather data in climate change era;

Precision Agriculture: Need for micro-climate data for optimal crop management;

Research & Education: Accessible platform for meteorological studies.

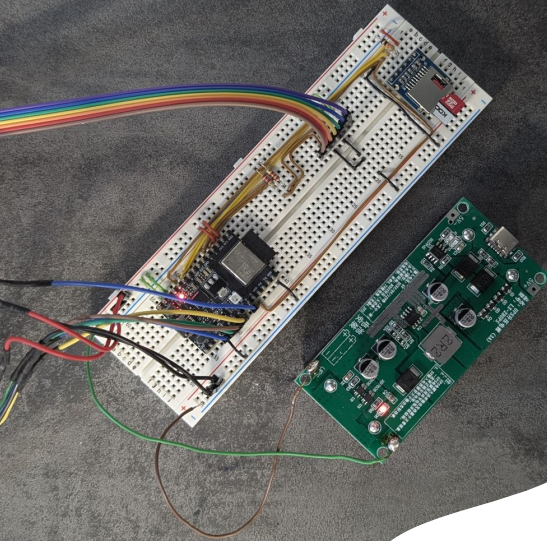
Technical Highlights

Multi-sensor Integration: BME280 + Custom made wind sensors in unified system;

Dual Data Strategy: Local SD card backup + Cloud MQTT publishing;

Configurable Connectivity: WiFi and MQTT settings via SD card;

Power Failure Resilient: Can be powered by batteries, solar panels and USB-C;



Introduction

Problem Statement

Commercial weather stations are expensive (\$500+), need for affordable alternatives;
Many areas lack adequate weather monitoring coverage

Our Propositions

Cost-Effective: Fraction of commercial station cost

Open Source: Fully customizable and extensible

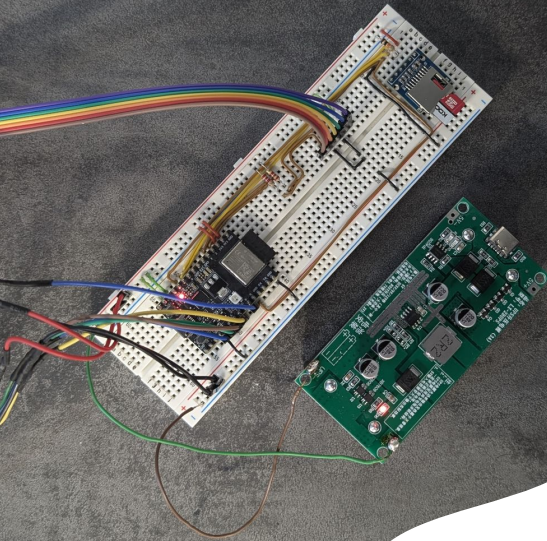
Scalable: Easy to deploy multiple units for coverage networks

Target Applications

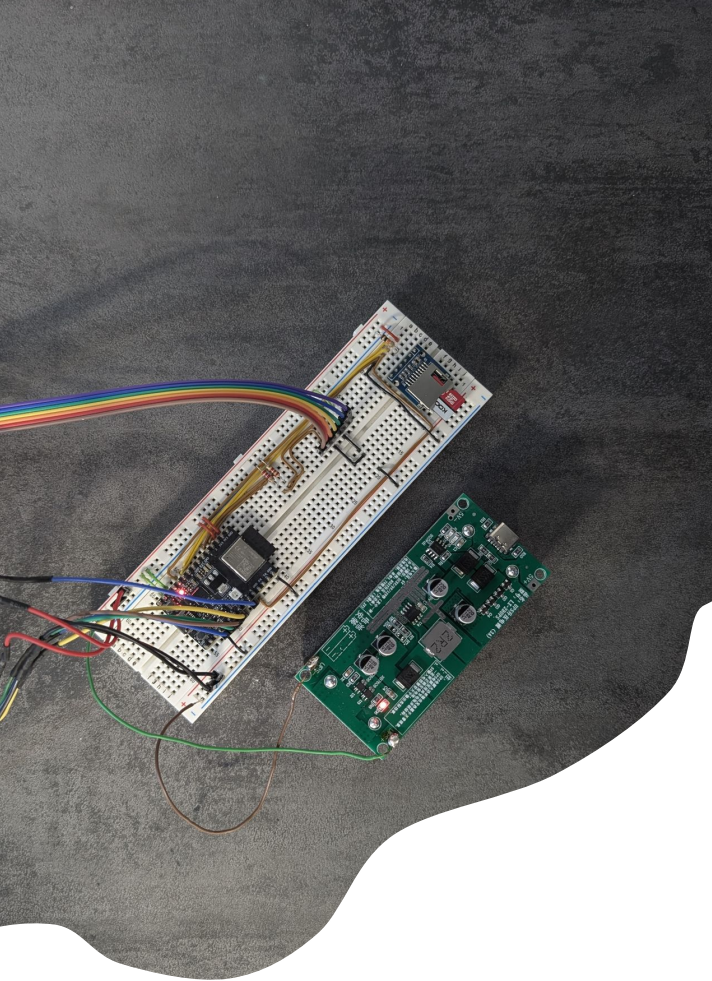
Agricultural monitoring systems

Weather enthusiast networks

IoT development learning platform



SW & HW Stack

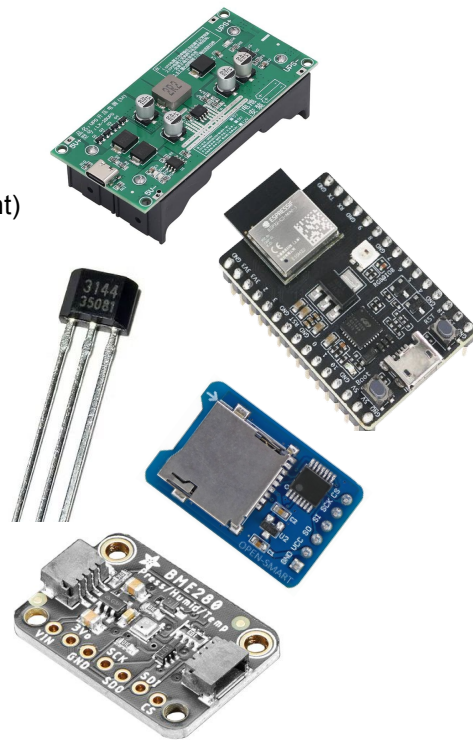


Hardware

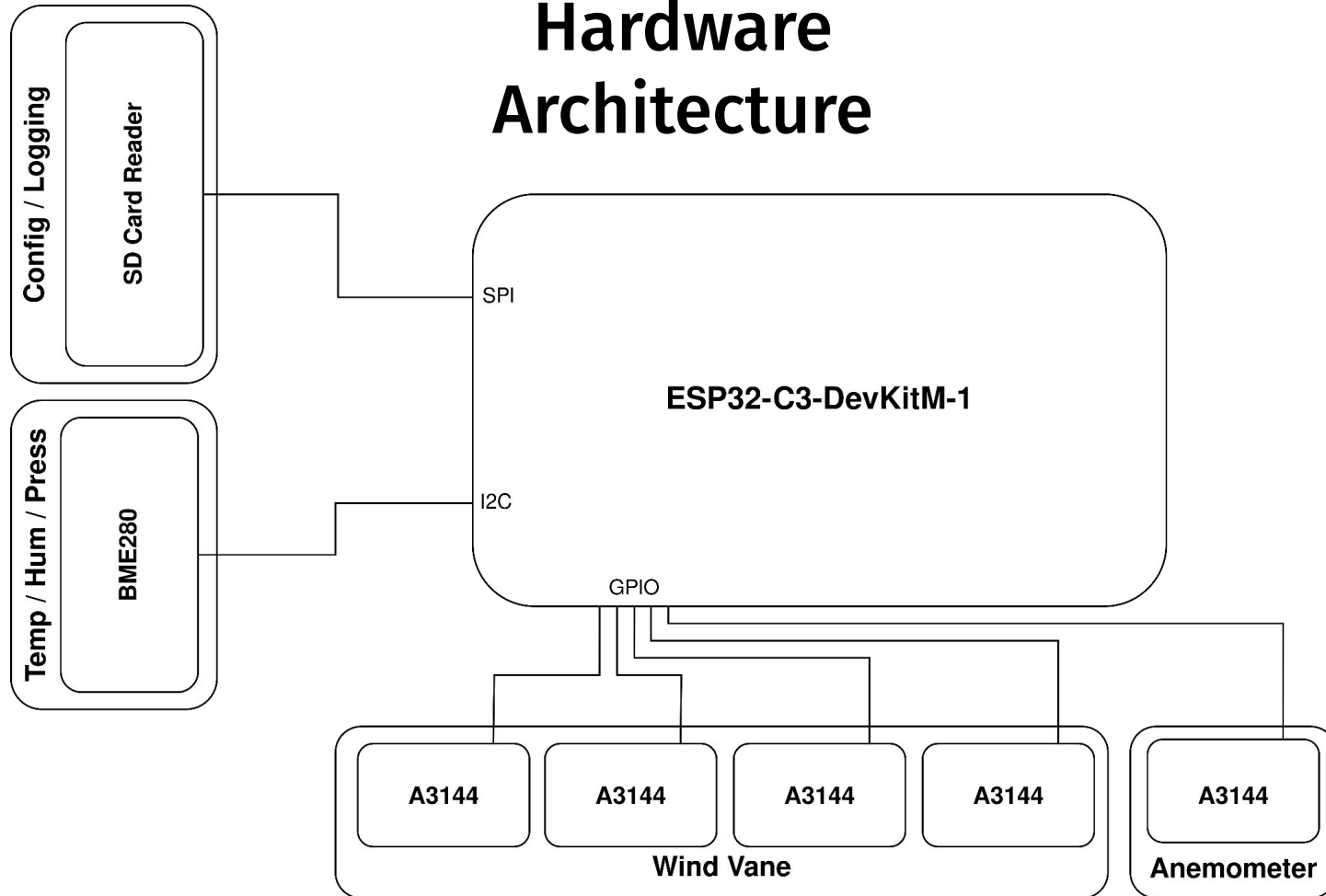
- ESP32-C3 SoC
- SPI SD Card Reader
- A3144 Hall Effect Sensor (latch & non latch variant)
- 18650 Lithium Battery UPS module
- BME280

Software

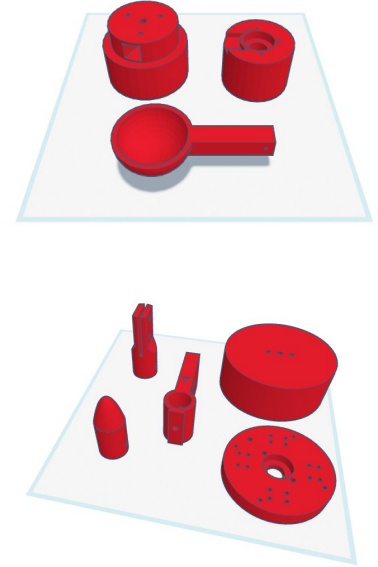
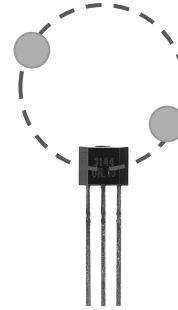
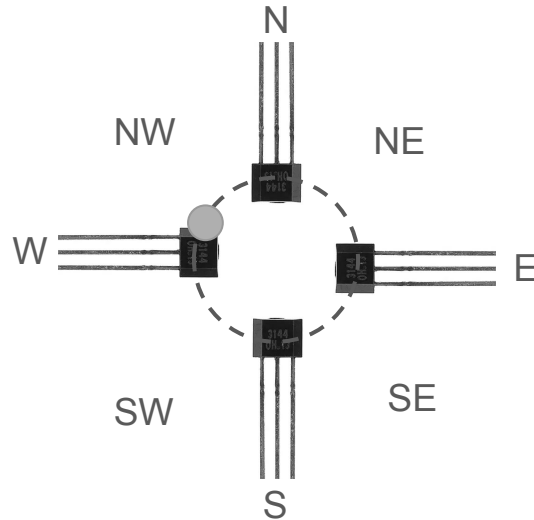
- ESP-IDF
- FreeRTOS
- MQTT
- WiFi
- Grafana



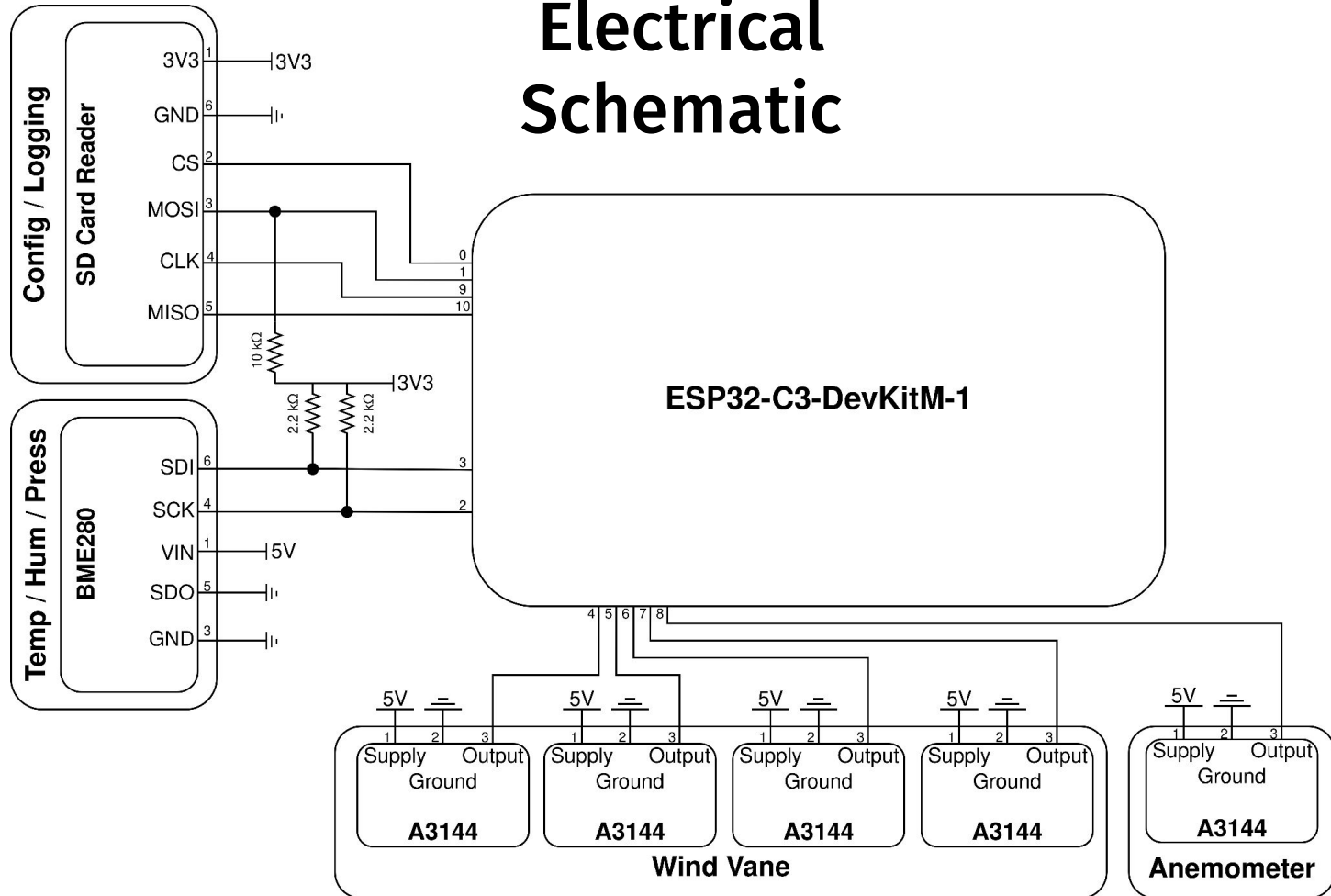
Hardware Architecture



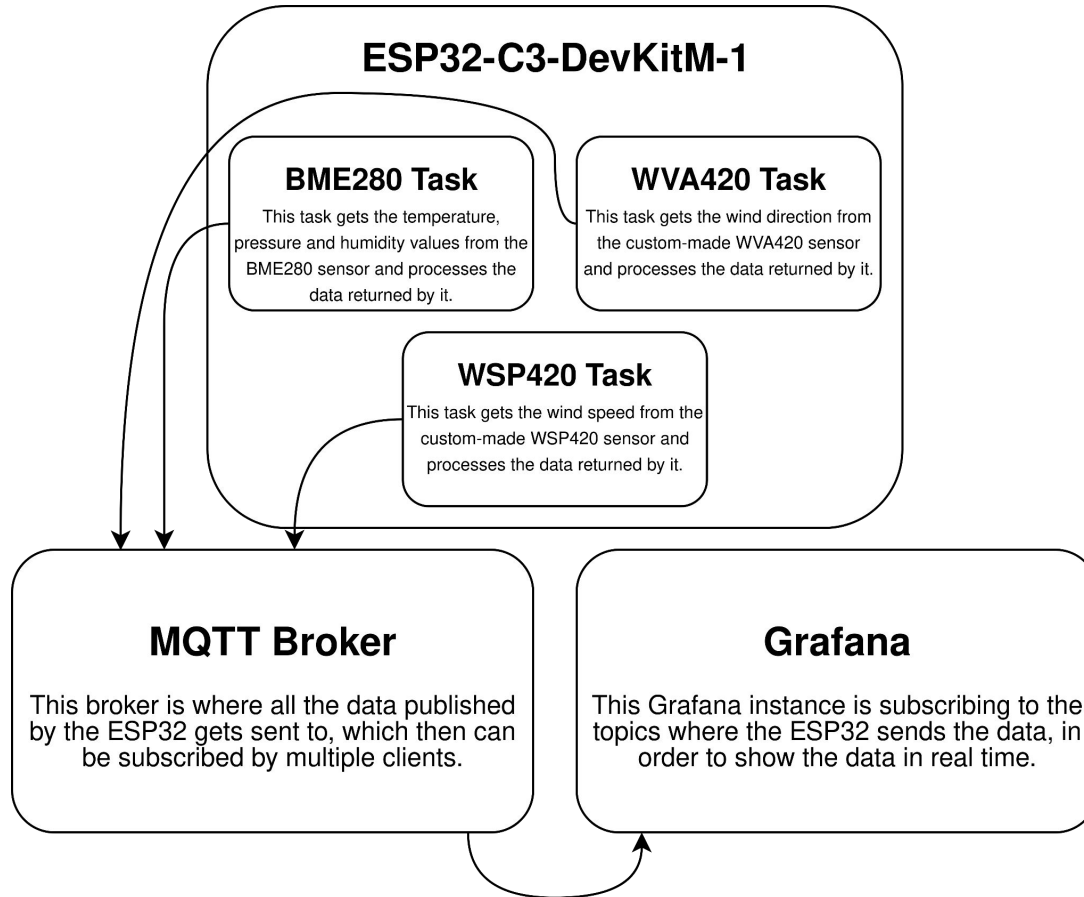
Custom-Made Sensors



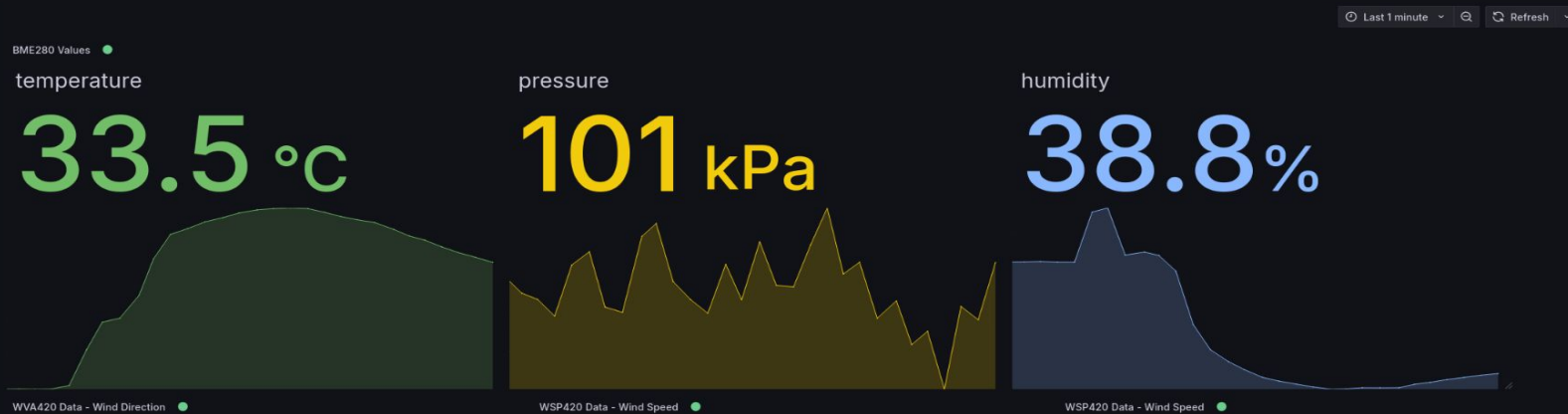
Electrical Schematic



Software Architecture



Demo



East

0.686 m/s

2.47 km/h

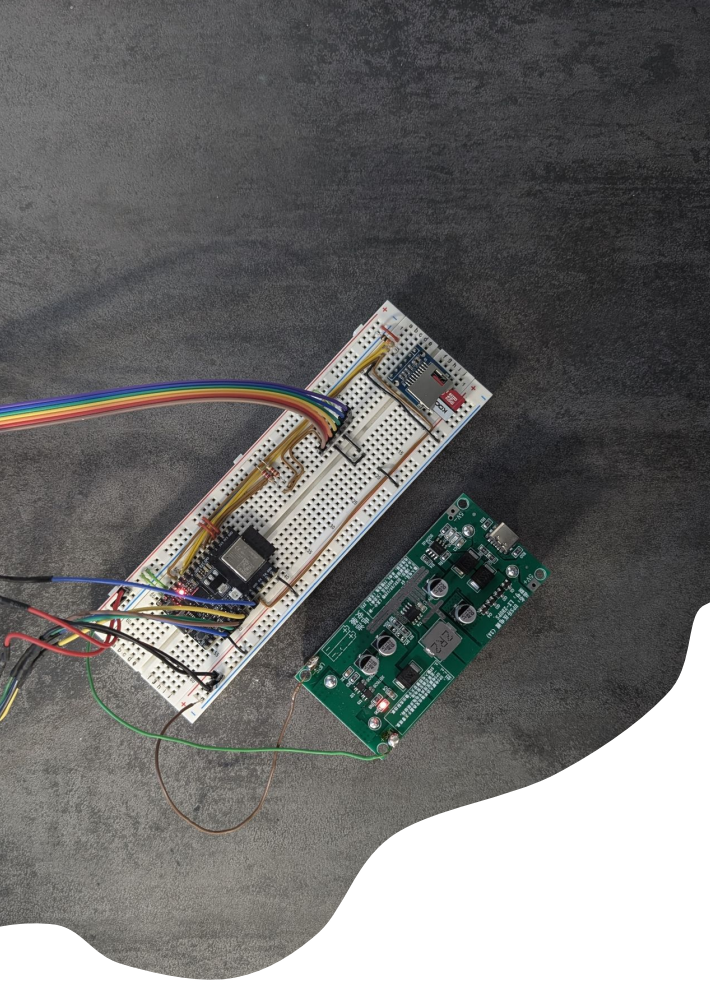
Battery Consumption

Power Draw: 0.35W

Current Draw: $0.35\text{W} / 5\text{V} = 0.07\text{A}$

Using two cells with 2200 mAh each, in parallel, we get 4400 mAh.

Battery life: $4400\text{ mAh} / 70\text{ mA} = 62.86\text{ hours}$



Future Work

Hardware

- Get the battery level reading from the UPS module
- Modularize the sensors (ATtiny)
- Improve the building quality of the custom-made sensors (weight, friction)

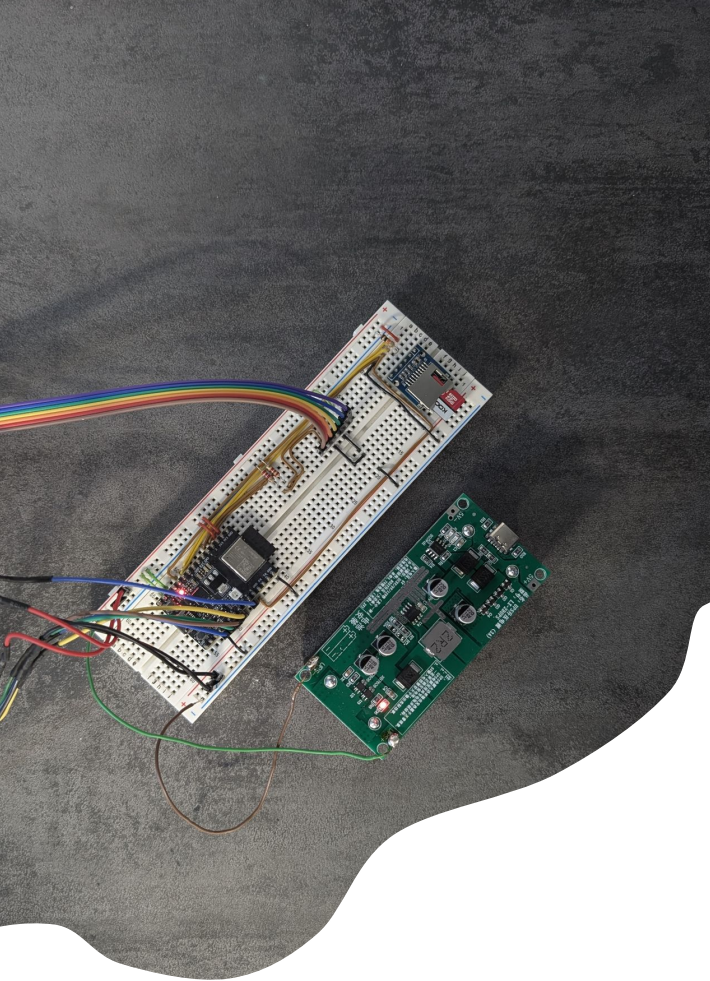
Software

- Energy management
- Wind direction sensor self-calibration
- Troubleshoot WiFi issues



Sources

- ESP-IDF Documentation
- Lab classes exercises
- Components datasheet



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universidade de aveiro
departamento de eletrónica,
telecomunicações e informática



Custom-made sensor: WSP420

